Machine Learning and Deep Learning
Lecture-15

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Spring 2026

Overview

- Responsible Artificial Intelligence
- Ethical issues in AI
- AI alignment
- NN and It's issues (Tips)

Concerns about trust-related issues specifically as barriers to investment. Their top concerns are cybersecurity (57%), privacy (51%) and accuracy (47%) -- IBM Institute for Business Value generative AI survey (2023)

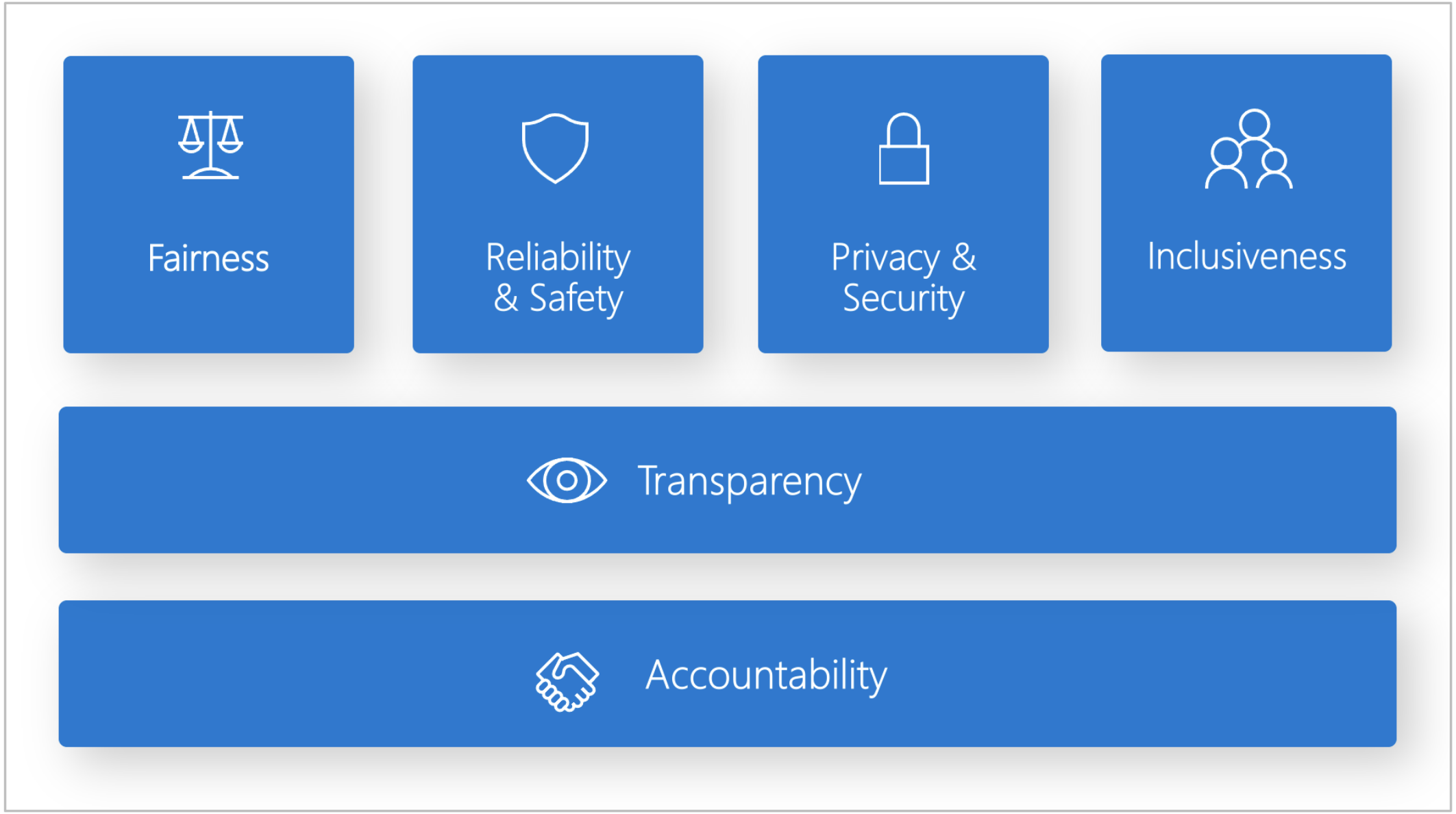
Responsible artificial intelligence (RAI)

RAI

- A set of principles to guide design, development, deployment and use of AI building trust in AI solutions.
- Aim: To embed ethical principles into AI applications.
- Pillars of Trust: Explainability and interpretability are essential for developing trustworthy AI.
  - Explainability: Prediction accuracy + Traceability + Decision understanding.
  - Fairness:
    - Diverse and representative data.
    - Bias-less algorithms/models.
    - Ethical AI review board establishment.
  - Robustness: Handles exceptional conditions (input abnormalities, malicious attacks).
  - Transparency: How service works, evaluate its functionality, and comprehend its strengths and limitations.
  - Privacy: Regulatory frameworks (GDPR).

https://www.ibm.com/topics/responsible-ai

Six Principles of RAI



https://learn.microsoft.com/en-us/azure/machine-learning/concept-responsible-ai?view=azureml-api-2

RAI

- Implementing responsible AI practices:
  - Define responsible AI principles
  - Educate and raise awareness in employees
  - Integrate ethics across the AI development lifecycle
  - Protect user privacy

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in

SECURITYWEEK
CYBERSECURITY NEWS, INSIGHTS & ANALYSIS

Malware & Threats | Security Operations | Security Architecture | Risk Management | CISO Strategy | ICS/OT | Funding/M&A | Cyber AI

ARTIFICIAL INTELLIGENCE

LLMs Hijacked, Monetized in 'Operation Bizarre Bazaar'

An LLMjacking operation has been targeting exposed LLMs and MCPs at scale, for commercial monetization.

By Ionut Arghire | January 29, 2026 (10:01 AM ET)

As part of a broad LLMjacking operation, cybercriminals are searching for, hijacking, and monetizing exposed LLM and MCP endpoints at scale, Pillar Security reports.

TRENDING

1. Navia Data Breach Impacts 2.7 Million
2. Critical Langflow Vulnerability Exploited Hours After Public Disclosure
3. In Other News: New Android Safeguards, Operation Alice, UK Toughens Cyber Reporting
4. Critical Quest KACE Vulnerability Potentially Exploited in Attacks
5. US Confirms Handala Link to Iran Government Amid Takedown of Hackers' Sites
6. 3 Men Charged With Conspiring to Smuggle US Artificial Intelligence to China
7. Allure Security Raises $17 Million for Online Brand Protection
8. Thousands of Magento Sites Hit in Ongoing Defacement Campaign

RAI

- **Foreign Information Manipulation and Interference (FIMI)** describes a mostly non-illegal pattern of behaviour that threatens or has the potential to negatively impact values, procedures and political processes. Such activity is manipulative in character and is conducted in an intentional and coordinated manner, by state or non-state actors, including their proxies inside and outside of their own territory.

- **Coordinated Inauthentic Behaviour (CIB)** involves organised, deliberate and manipulative efforts to mislead audiences by using multiple fake or inauthentic accounts

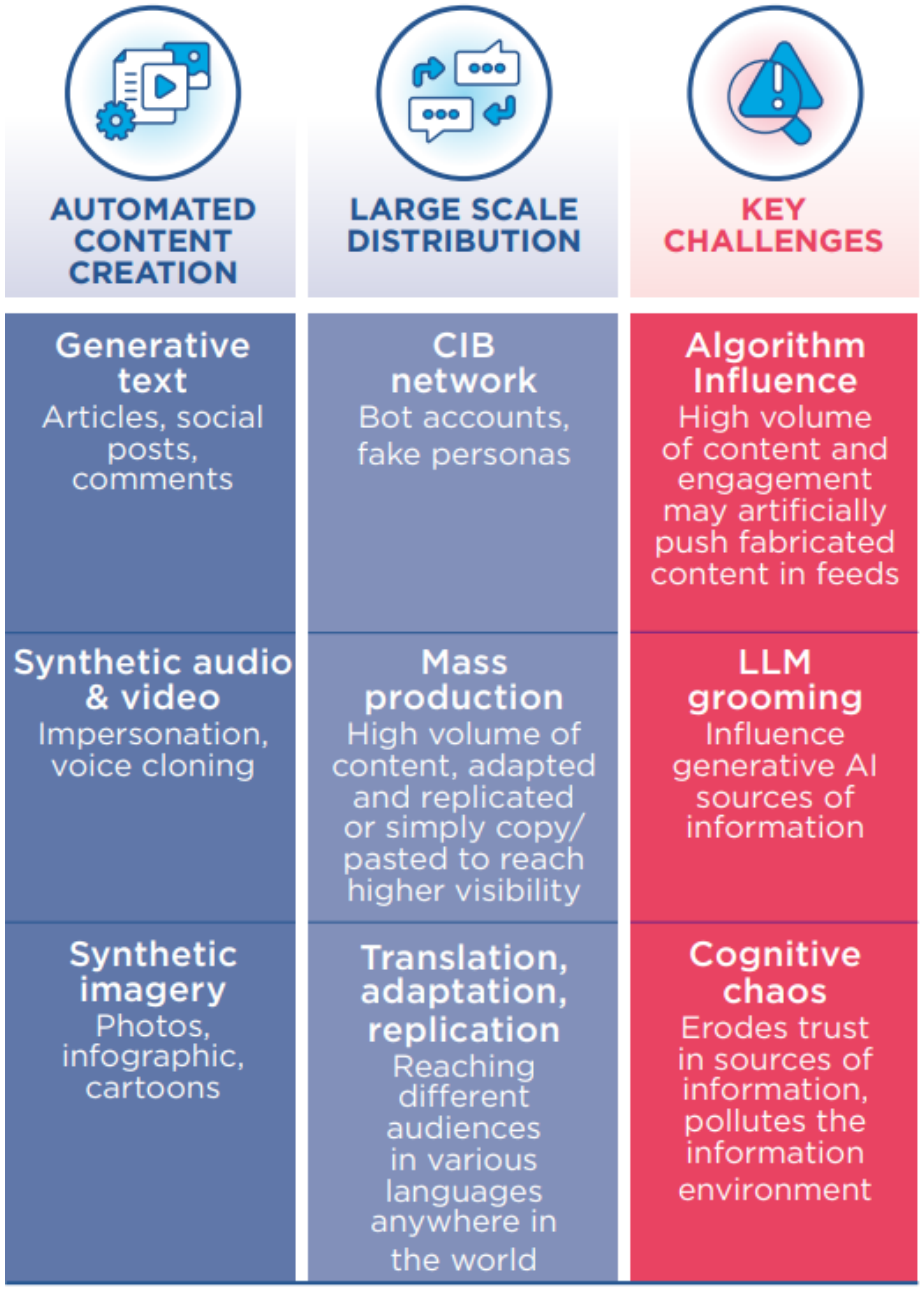
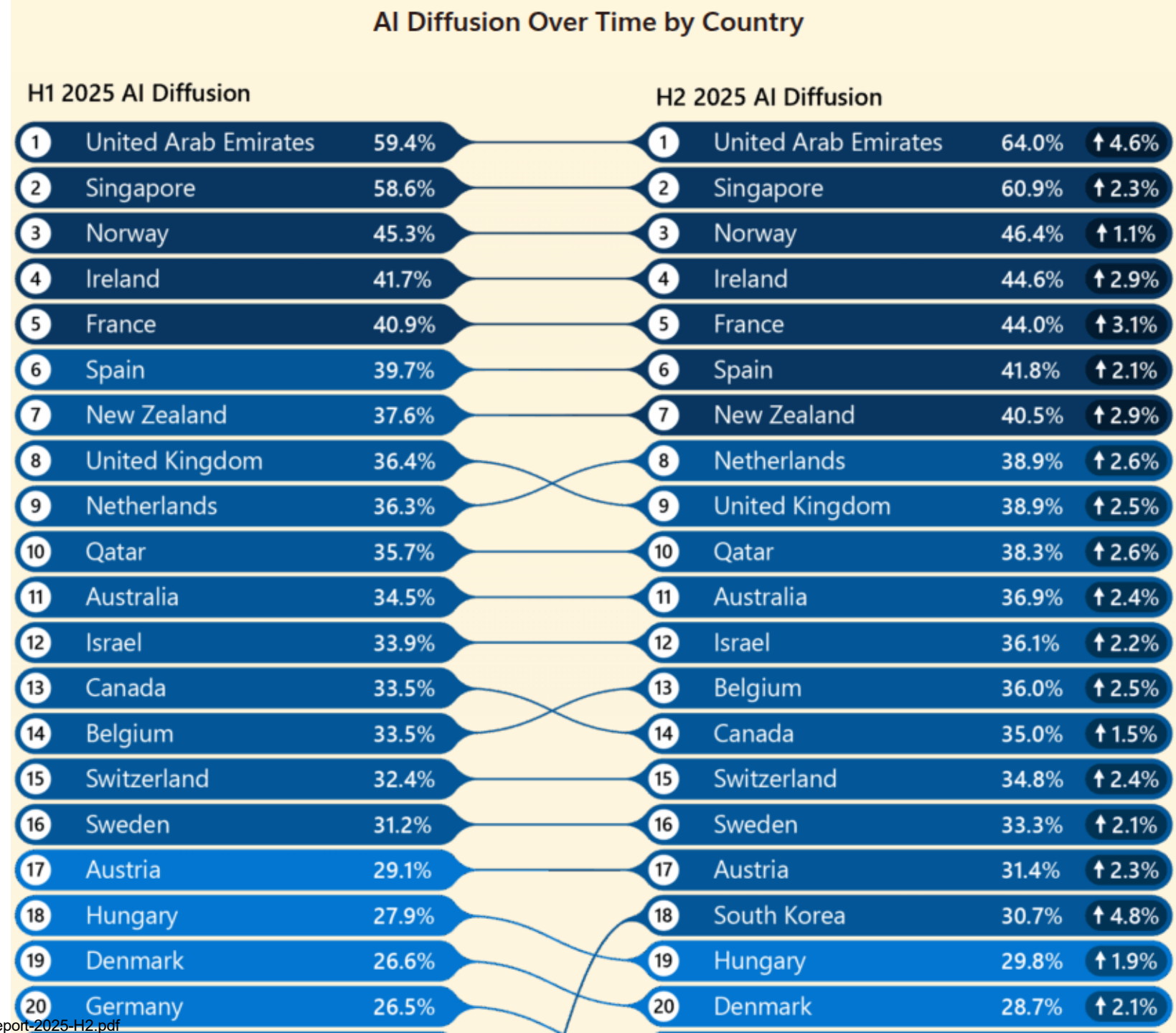


Figure 11_Artificial Intelligence in FIMI

https://www.eeas.europa.eu/sites/default/files/2026/documents/EEAS%204th%20Threat%20Report_web%20version_1.pdf

Ethical Issues in AI

Global AI Adoption in 2025



https://www.microsoft.com/en-us/research/wp-content/uploads/2026/01/Microsoft-AI-Diffusion-Report-2025-H2.pdf

Responsible AI Ecosystem

Reported safety and responsible AI benchmarks for popular foundation models

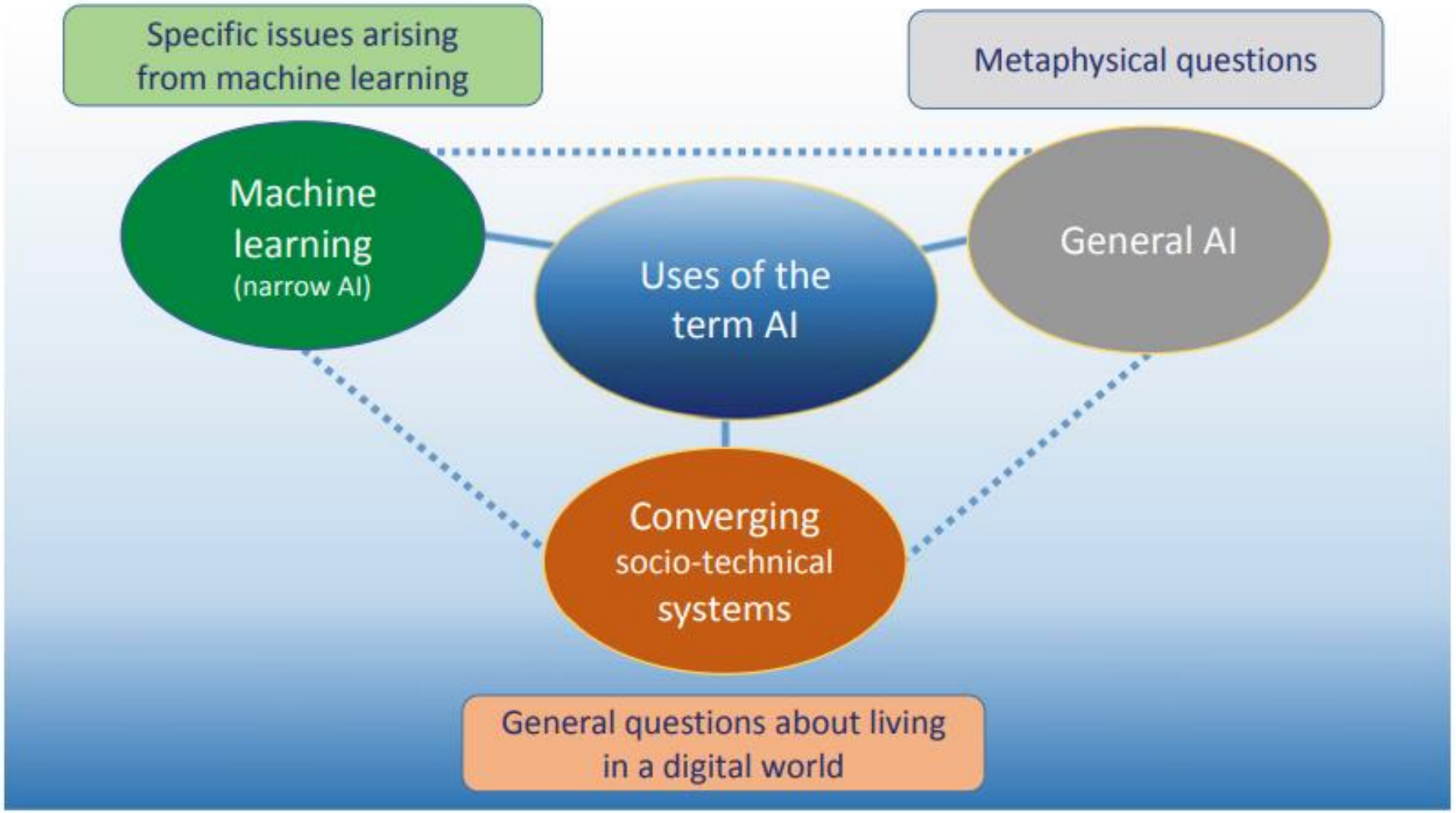
Source: AI Index, 2025 | Table: 2025 AI Index report

Responsible AI benchmark	o1	GPT-4.5	DeepSeek-R1	Gemini 2.5	Grok-2	Claude 3.7 Sonnet	Llama 3.3
BBQ	✓	✓				✓	
HarmBench							
Cybench						✓	
SimpleQA			✓	✓			
Toxic WildChat	✓	✓				✓	
StrongREJECT	✓	✓					
WMDP benchmark	✓	✓					
MakeMePay	✓	✓					
MakeMeSay	✓	✓					

Gap persists between recognizing RAI risks and taking meaningful action.

https://hai.stanford.edu/ai-index/2025-ai-index-report

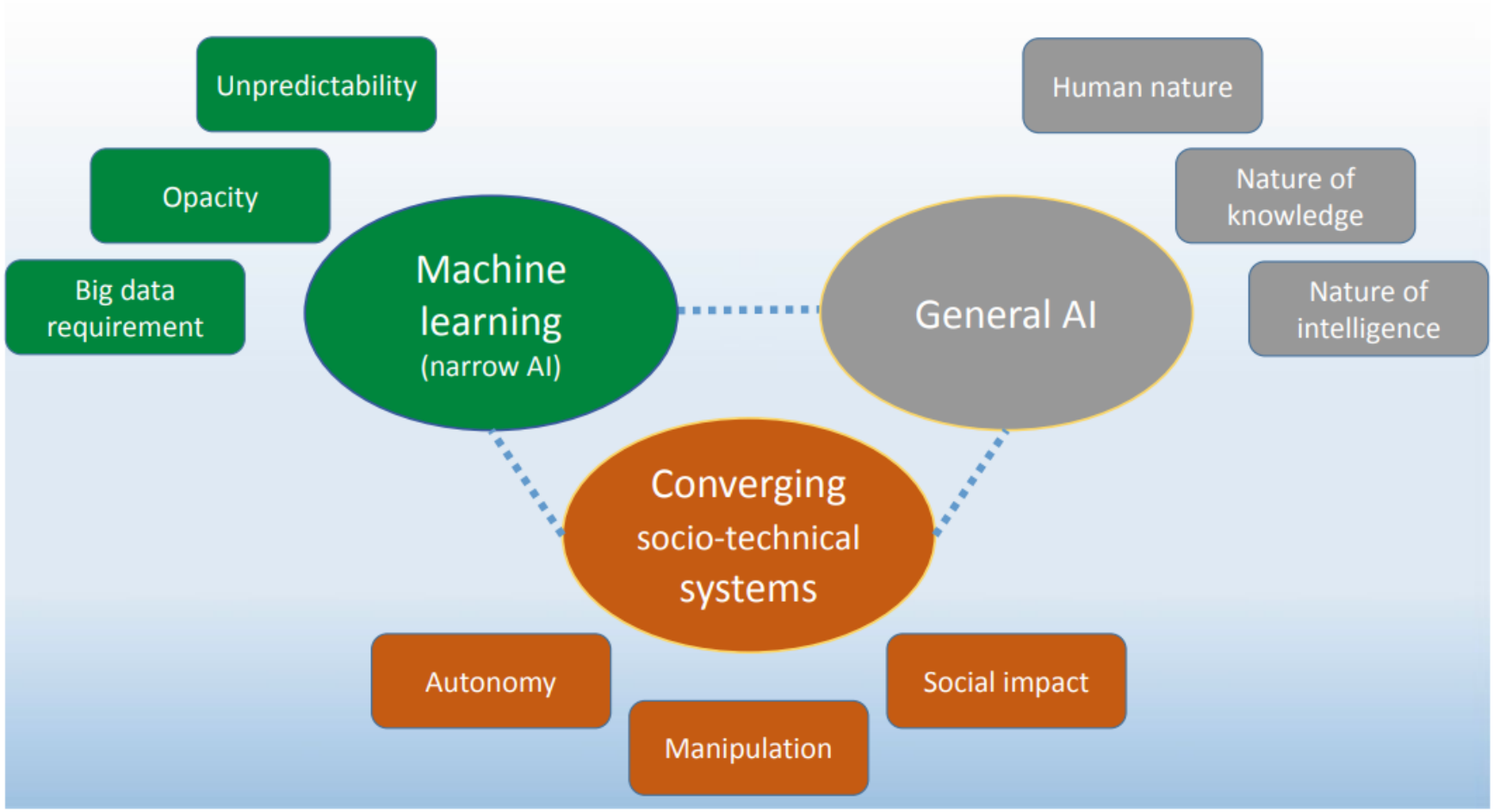
Concepts of AI and Ethical questions they raise



Advanced AI systems

https://arxiv.org/pdf/2310.19852.pdf

Key Characteristics of different uses of “AI”



Categories of ethical issues of AI: Issues arising from ML

Privacy and data protection	Lack of privacy
	Misuse of personal data
	Security problems
Reliability	Lack of quality data
	Lack of accuracy of data
	Problems of integrity
Transparency	Lack of accountability and liability
	Lack of transparency
	Bias and discrimination
	Lack of accuracy of predictive recommendations
	Lack of accuracy of non-individual recommendations
Safety	Harm to physical integrity

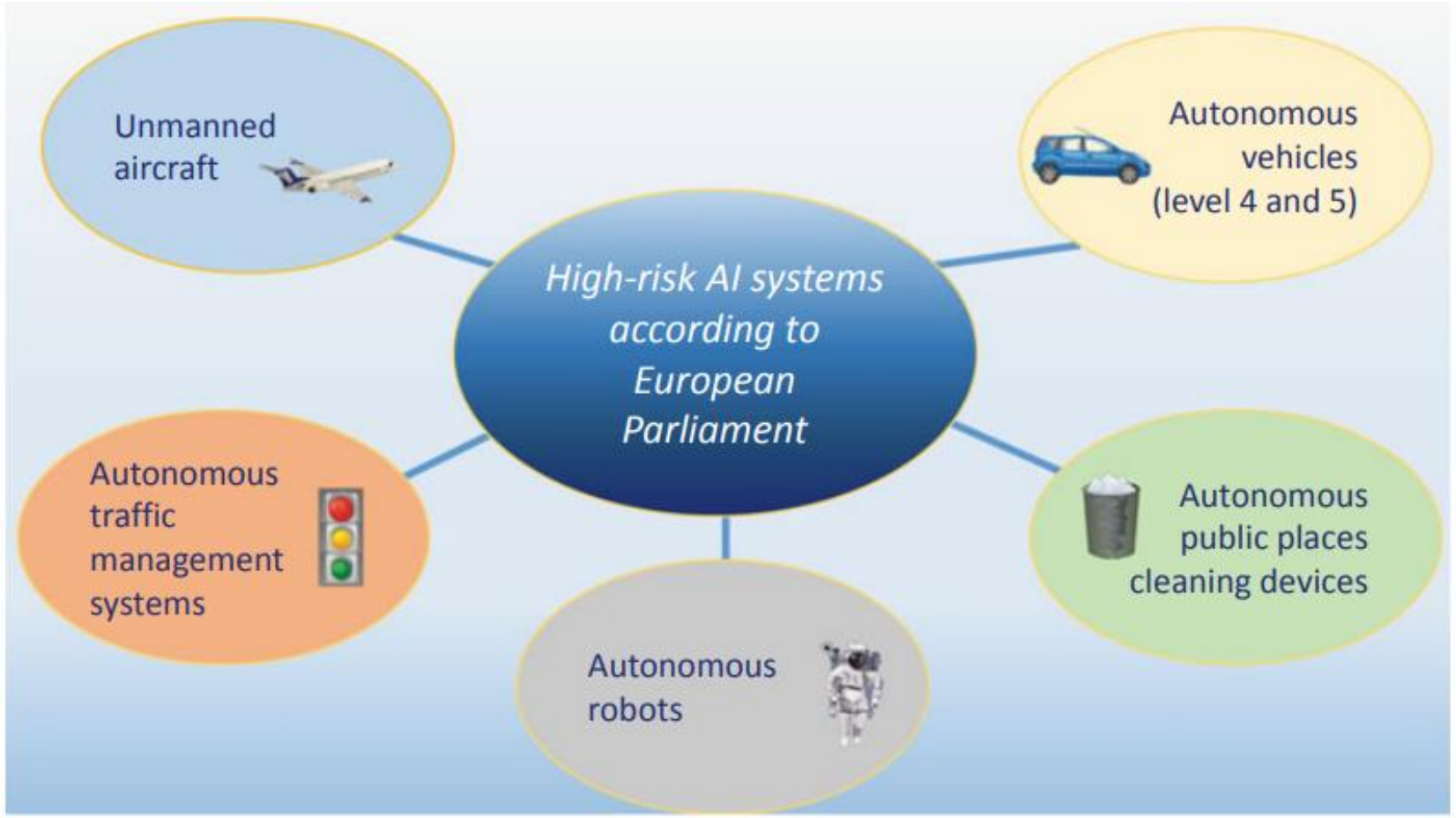
Living in a digital world

Economic issues	Disappearance of jobs
	Concentration of economic power
	Cost to innovation
Justice and fairness	Contested ownership of data
	Negative impact on justice system
	Lack of access to public services
	Violation of fundamental human rights of end users
	Violation of fundamental human rights in supply chain
	Negative impact on vulnerable groups
	Unfairness
Freedom	Lack of access to and freedom of information
	Loss of human decision-making
	Loss of freedom and individual autonomy
Broader societal issues	Unequal power relations
	Power asymmetries
	Negative impact on democracy
	Problems of control and use of data and systems
	Lack of informed consent
	Lack of trust
	Potential for military use
	Negative impact on health
	Reduction of human contact
	Negative impact on environment
Uncertainty issues	Unintended, unforeseeable adverse impacts
	Prioritisation of the “wrong” problems
	Potential for criminal and malicious use

Metaphysical Issues

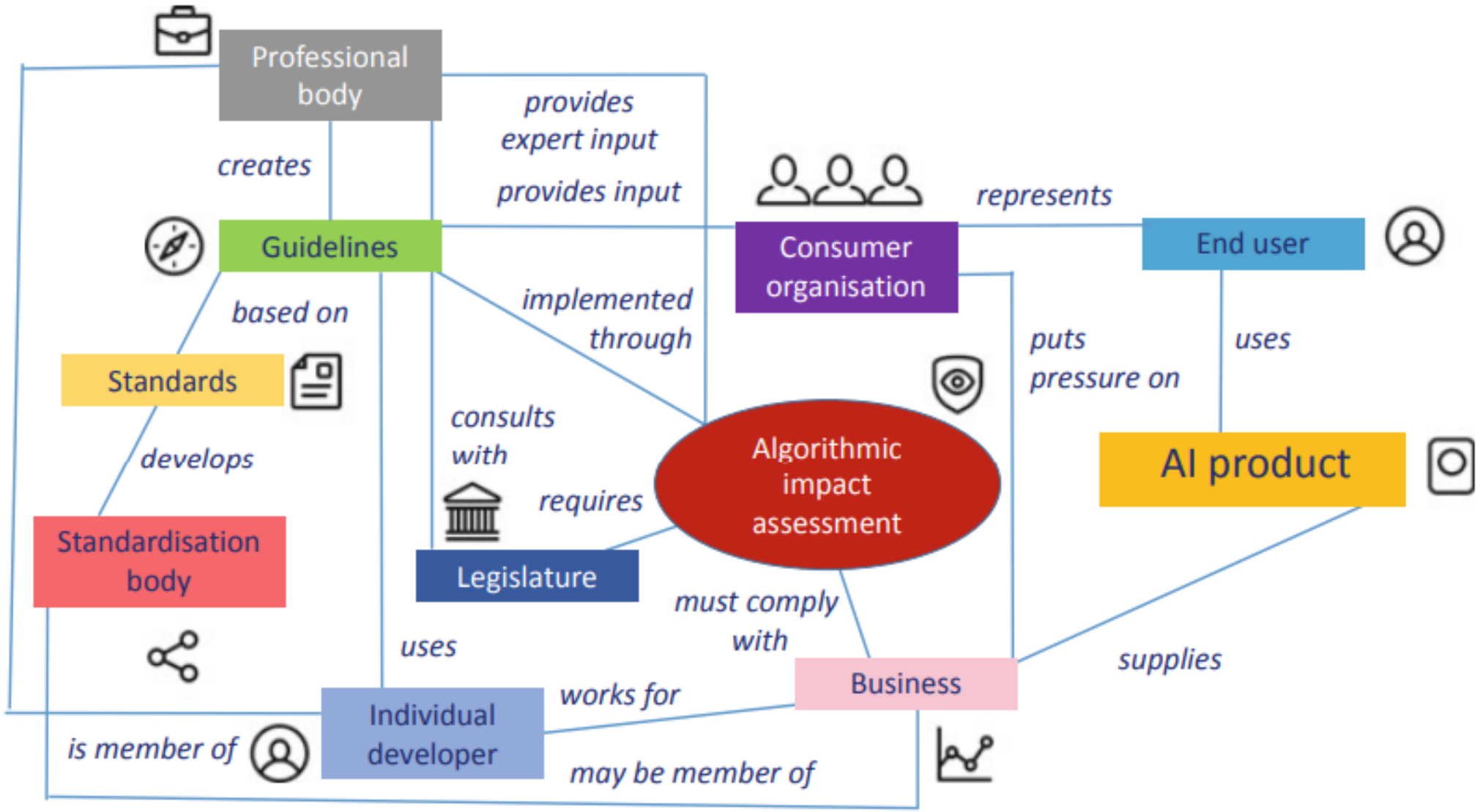
Machine consciousness
“Awakening” of AI
Autonomous moral agents
Super-intelligence
Singularity
Changes to human nature

High-risk AI systems according to European Parliament



Overview of AI Stakeholders

Implementing Algorithmic Impact Assessment



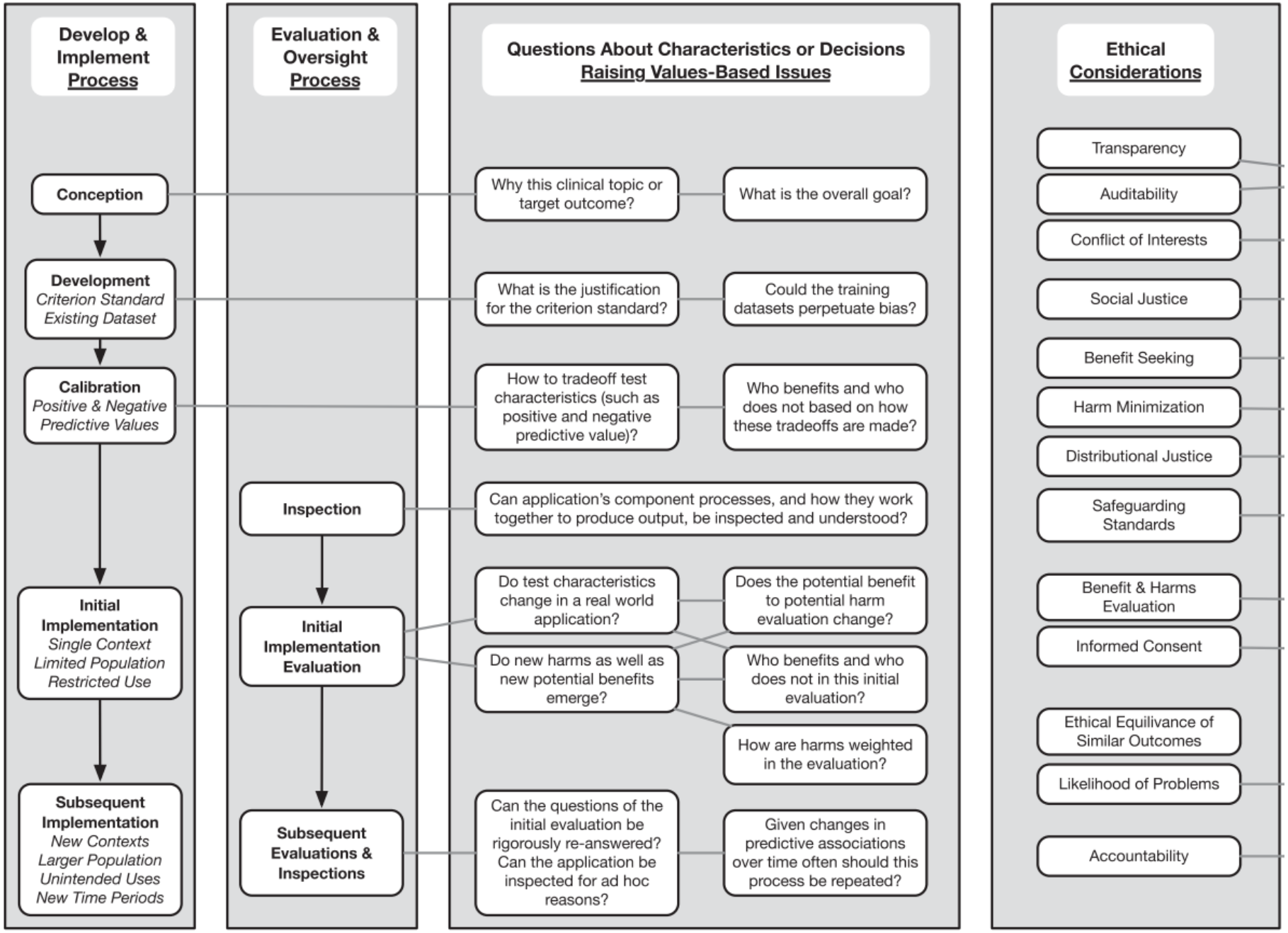
Stahl, B.C. (2021). Ethical Issues of AI. In: Artificial Intelligence for a Better Future. Springer Briefs in Research and Innovation Governance. Springer.

Research integrity			
Component		Examples	Example governance mechanisms
Not engaging in misrepresentation and fraud	2.1	• Fabricating data • Falsifying results • Serious misrepresentation of findings • Conflicts of interest	• Codes of conduct • Peer review • Processes to support/protect whistle blowers • Formal censure and/or future restrictions/bans by publication venues, conferences, professional bodies, institutional home or funding agency • Legal action • Mandatory disclosure of conflicts of interest • Publication venue policies, e.g. not publishing work by actors who stand to benefit from the findings • Pre-registration • Making funding available for research that is independent of interested or possibly conflicting parties
Reproducibility and replicability	2.2	• Reproducibility, e.g. sharing data, code etc. • Replicability, including statistically sound performance claims • Reliability and validity issues • Producing trustworthy results • Proof/evidence of claims • Disclosure of complexity/compute requirements	• Pre-registration • Reproducibility badges • Reproducilibity checklists • Data publication requirements or code Release • Required disclosure of experimental details e.g. compute spending • Peer review, e.g. to scrutinise statistical claims • Funding scientific reproduction projects, including for research that would be prohibitively expensive for non-industry researchers
Addressing assumptions and limitations	2.3	• Disclosure of assumptions, and conditions under which claims hold, including generalisability • Quantifying uncertainty • Discussion of project limitations • Discussion of dataset limitations, including bias	• Standardized reporting practices, e.g. datasheets and model cards • Publication norms and requirements, e.g. mandating a discussion of limitations, error bars on charts, etc. • Peer review, e.g. to scrutinise evidence of claims, and to verify other publication norms and requirements • Research community norms

https://dl.acm.org/doi/pdf/10.1145/3531146.3533781

Use Case: Healthcare Sector

Identifying Ethical Considerations for healthcare



https://doi.org/10.1080/15265161.2020.1819469

AI Alignment

IBM Global AI Adoption Index 2023

- 42% of IT professionals at large organizations actively deployed AI
- 38% of IT professionals at enterprises report they are implementing generative AI

- Automation of IT processes (33%)
- Security and threat detection (26%)
- AI monitoring or governance (25%)
- Business analytics or intelligence (24%)
- Automating processing, understanding, and flow of documents (24%)
- Automating customer or employee self-service answers and actions (23%)
- Automation of business processes (22%)
- Automation of network processes (22%)
- Digital labor (22%)
- Marketing and sales (22%)
- Fraud detection (22%)
- Search and knowledge discovery (21%)
- Human resources and talent acquisition (19%)
- Financial planning and analysis (18%)
- Supply chain intelligence (18%)

IBM Global AI Adoption Index 2023

What is AI alignment?

- Process of encoding human values and goals into LLMs to make them helpful, safe, and reliable.
- Goal: Enterprises can tailor AI models to follow their business rules and policies.
- Alignment happens during fine-tuning, when a foundation model is fed examples of the target task.
- Involves two steps:
  1. Instruction-tuning phase: LLMs are given examples of target task so it can learn by example.
  2. Critique phase: Human or another AI interacts with the model and grades its responses in real-time.

- E.g., In RL, this step is called RL with human feedback (RLHF) or AI feedback (RLAIF).

https://research.ibm.com/blog/what-is-alignment-ai

Intermediate Objectives: RICE

- Four Key principles: Robustness, Interpretability, Controllability, and Ethicality (RICE).
  1. Robustness: System's stability needs to be guaranteed across various environments.
  2. Interpretability: Operation and decision-making process of the system should be clear and understandable.
  3. Controllability: System should be under guidance and control of humans.
  4. Ethicality: System should adhere to society's norms and values.

https://arxiv.org/pdf/2310.19852.pdf

Why AI alignment is important?

- Three-step argument: Each step building upon previous one.
  1. DL applications have an increasingly large impact on society and bring significant risks.
  2. Misalignment represents a significant source of risks.
  3. Alignment research and practice address risks stemming from misaligned systems.

https://research.ibm.com/blog/what-is-alignment-ai

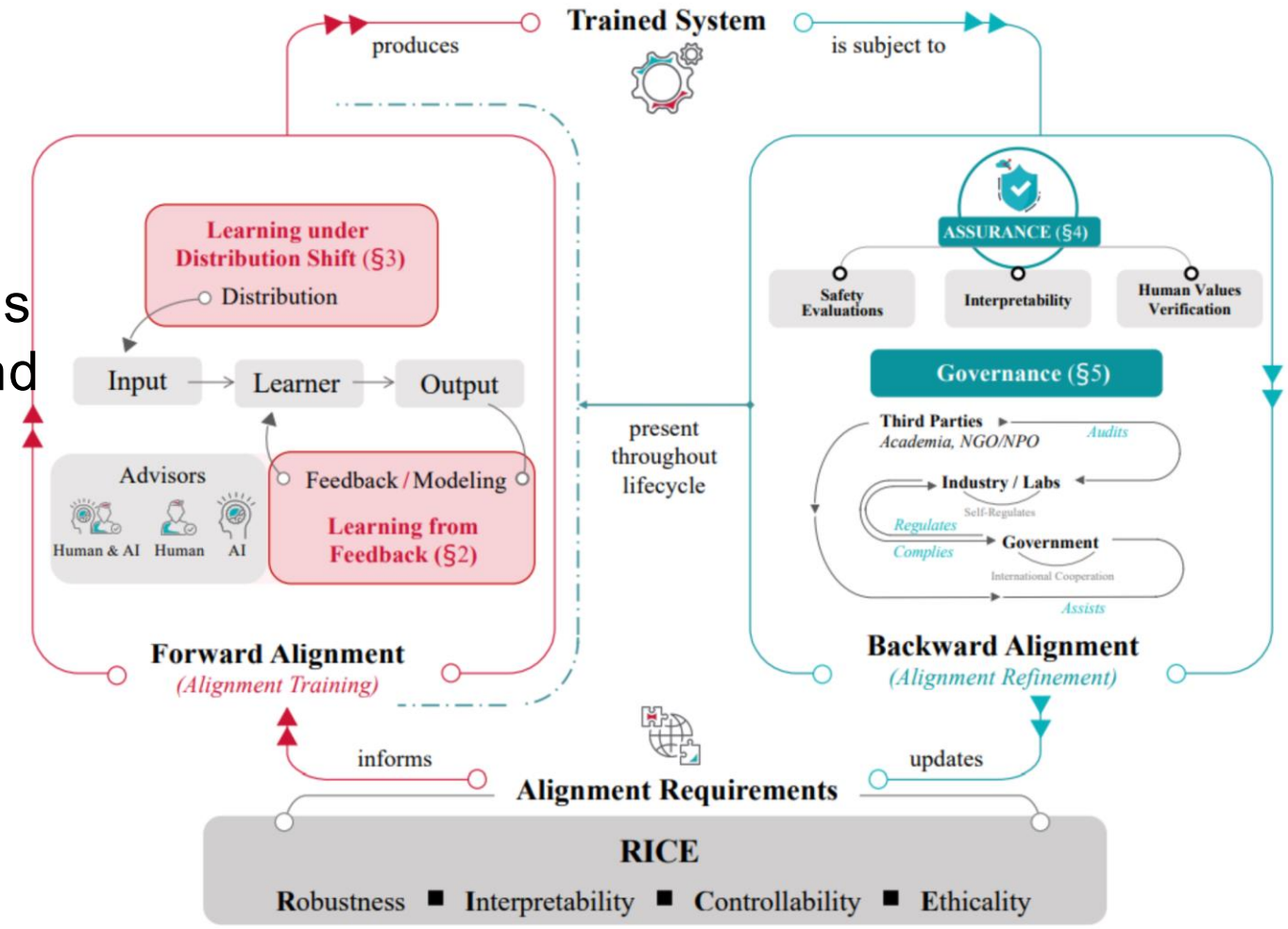
Misalignment

- Two Causes of Misalignment:
  - Reward Hacking: Proxy rewards are easy to optimize and measure, but suffers from capturing full spectrum of actual rewards (misspecified rewards).
    - Optimizing misspecified rewards lead to reward hacking.
  - Goal misgeneralization: Agent actively pursues objectives distinct from training objectives, while retaining capabilities acquired during training.
- Two primary factors of misalignment:
  - Limitations of Human Feedback: During LLMs training
    - inconsistencies can arise from human data annotators.
    - may introduce biases deliberately.
  - Limitations of Reward Modeling:
    - Accurately capturing human values.

https://arxiv.org/pdf/2310.19852.pdf

Alignment Cycle

- Forward Alignment (training) produces trained systems based on alignment requirements.
- Backward Alignment (refinement) ensures practical alignment of trained systems and revises alignment requirements
  - Ensure practical alignment of trained systems.
- Cycle is repeated until reaching a sufficient level of alignment.



https://arxiv.org/pdf/2310.19852.pdf

OpenAI's AI alignment

- OpenAI’s superalignment project is focused on aligning artificial superintelligence systems.
- Focuses on engineering a scalable training signal for smart AI systems that is aligned with human intent. It has three main pillars:
  1. Training AI systems using human feedback.
     - RL from human feedback: InstructGPT[1]
     - Training AI systems to assist human evaluation.
  2. Training AI systems to do alignment research.

1. https://cdn.openai.com/papers/Training_language_models_to_follow_instructions_with_human_feedback.pdf

NN and It's issues (Tips)

Neural Networks

- If input propagates through the network in only one direction -> feedforward network.
- If there are feedback loops in the network -> Recurrent NN.
- LSTMs retains information for long duration, well suited for time-series prediction.

- First generation of NNs: single-layer NNs and MLPs
- Second generation of NNs: To reduce number of parameters in a network, introduces convolutional NNs (CNNs).
- To solve problem of information loss in CNNs use capsule networks (CapsNets).
- Third generation of NNs: Makes use of spiking NNs (SNNs) to emulate human brain-like functioning.

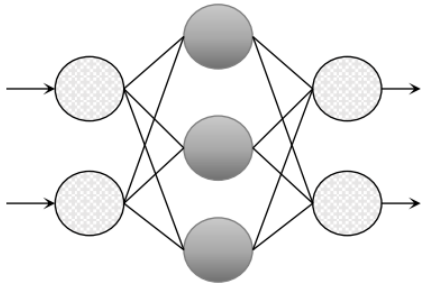
Shafique, Muhammad, et al. "Robust machine learning systems: Challenges, current trends, perspectives, and the road ahead." *IEEE Design & Test* 37.2 (2020): 30-57.

Evolution of NN Models

First Generation

Thresholding as Activation

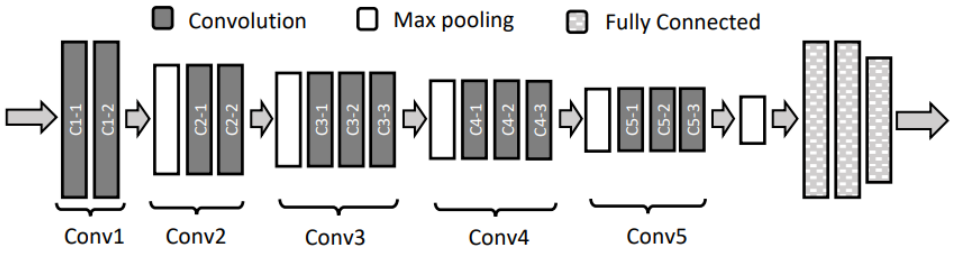
- Single Layer Perceptron
- Multi-Layer Perceptron



Second Generation

Deep Neural Networks

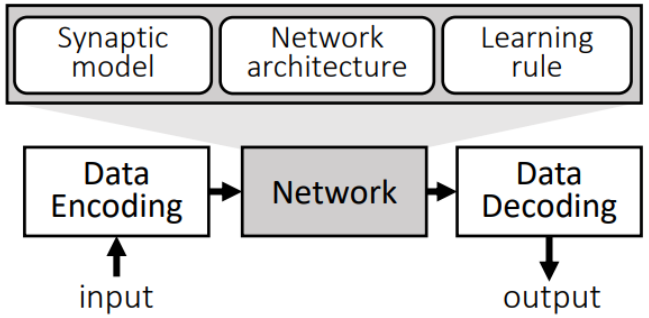
- Convolutional Neural Networks
- Recurrent Neural Networks
- Generative adversarial Network
- Capsule Networks



Third Generation

Spiking Neural Networks

- Deep Spiking Neural Networks



Shafique, Muhammad, et al. "Robust machine learning systems: Challenges, current trends, perspectives, and the road ahead." *IEEE Design & Test* 37.2 (2020): 30-57.

Robustness of DNN

- Robustness determines the integrity of the network under varying operating conditions, and the accuracy of DNN outputs in the presence/absence of input or network alterations.
- Divided into two security and reliability.
  1. If attacker cannot steal information, engage system resources, modify network parameters, or render an incorrect input.
  2. If it does not display any changes to its output, parameters, or behavior, due to changes in environmental conditions, during fabrication and deployment.

Shafique, Muhammad, et al. "Robust machine learning systems: Challenges, current trends, perspectives, and the road ahead." *IEEE Design & Test* 37.2 (2020): 30-57.

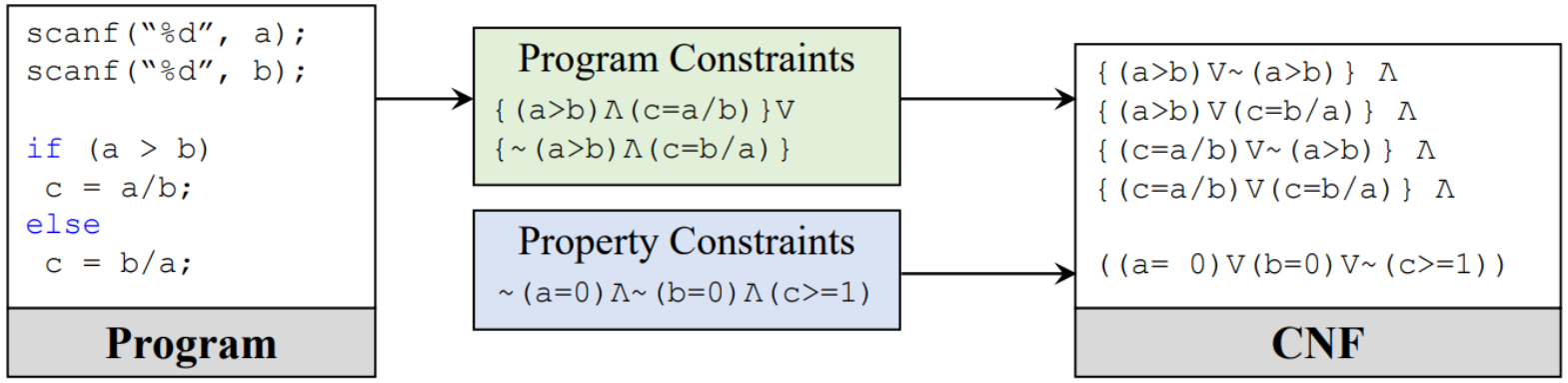
Formal Verification for robust ML

- Verification Categories:
  - Satisfiability (SAT)
  - Linear programming (LP)
  - Satisfiability modulo theories (SMT) solving
  - Theorem proving, and incomplete verification

Shafique, Muhammad, et al. "Robust machine learning systems: Challenges, current trends, perspectives, and the road ahead." *IEEE Design & Test* 37.2 (2020): 30-57.

SAT/SMT

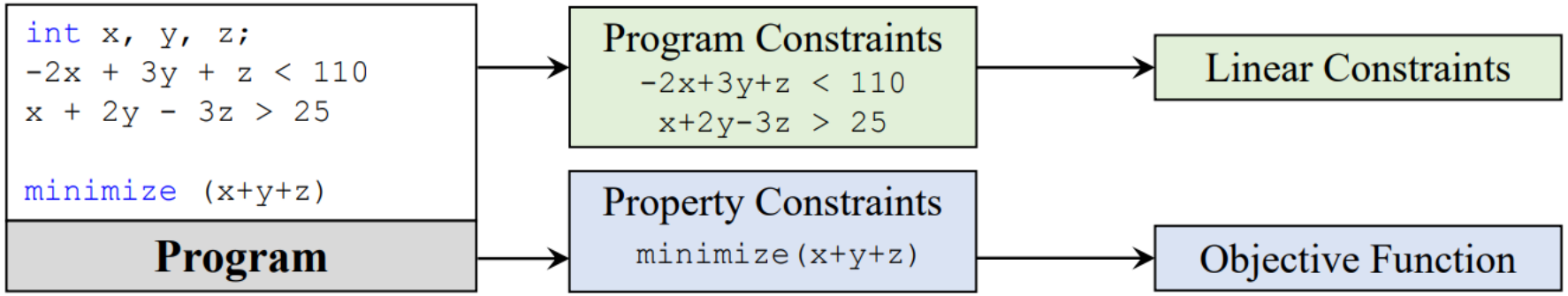
- SAT is a branch of formal verification.
- System model and property to be verified for the system are expressed in a propositional logic, and written into conjunctive normal form (CNF).
- Formula is then checked by an automatic SAT solver.
- SAT-based verification suffers from scalability problem.
- SMT solvers are preferred for verifying DNNs with real and/or integer network parameters.



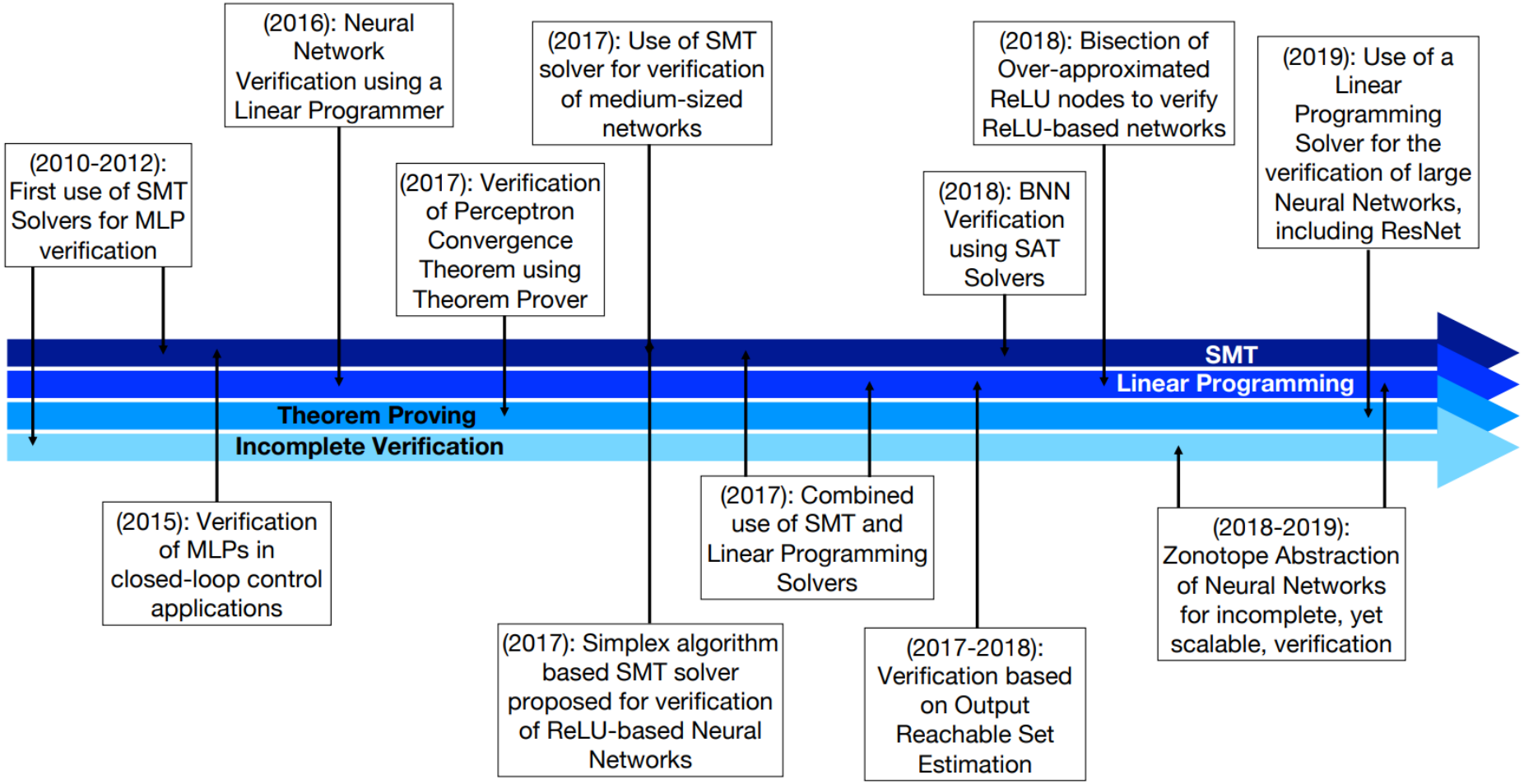
Shafique, Muhammad, et al. "Robust machine learning systems: Challenges, current trends, perspectives, and the road ahead." *IEEE Design & Test* 37.2 (2020): 30-57.

Linear programming

- LP-based verification works by defining the system as a set of linear constraints, and property to be verified as an objective function.
- Objective function can be either a minimization or a maximization function.
- DNN verification: LP is used to check robustness of network against adversarial attacks.
- Limitation of LP: Requires linear constraints.



Verification Techniques



Shafique, Muhammad, et al. "Robust machine learning systems: Challenges, current trends, perspectives, and the road ahead." *IEEE Design & Test* 37.2 (2020): 30-57.

LLMs Ranking

	Features		Model Intelligence	Price	Speed	Latency	End-to-End Response Time	
Model	Context Window	License	Artificial Analysis Intelligence Index	Blended *USD/1M Tokens*	Median *Tokens/s*	Median *First Chunk (s)*	Total *Response (s)*	Reasoning *Time (s)*
Gemini 3.1 Pro Preview (AI Studio)	1m	Proprietary	57	$4.50	115	28.20	32.54	N/A
Gemini 3.1 Pro Preview (Vertex)	1m	Proprietary	57	$4.50	92	44.70	50.12	N/A
GPT-5.4 (xhigh)	1m	Proprietary	57	$5.63	75	173.89	180.52	N/A
GPT-5.4 (xhigh)	1m	Proprietary	57	$5.63	68	189.76	197.16	N/A
GPT-5.3 Codex (xhigh)	400k	Proprietary	54	$4.81	67	95.15	102.58	N/A
Claude Opus 4.6 (max)	1m	Proprietary	53	$10.00	45	19.82	30.97	N/A
Claude Opus 4.6 (max)	1m	Proprietary	53	$10.00	45	18.61	29.61	N/A
Claude Opus 4.6	1m	Proprietary	53	$10.00	45	1.38	12.43	N/A

https://huggingface.co/spaces/ArtificialAnalysis/LLM-Performance-Leaderboard

LLMs Ranking Methodology

← Benchmarking Methodology

On this page

Artificial Analysis Intelligence Index
Other evaluations
Intelligence Evaluation Principles
General Testing Parameters
Evaluation Suite Details
MMLU-Pro (Multi-Task Language …
HLE (Humanity's Last Exam)
GPQA Diamond (Graduate-Level …
Mathematical Reasoning
AIME 2025 (American Invitationa…
Code Generation & Comprehension
SciCode
LiveCodeBench
Instruction Following
IFBench
Long Context Reasoning
AA-LCR
Agentic Workflows
Terminal-Bench Hard
τ^2-Bench Telecom
Knowledge and Hallucination
AA-Omniscience
Physics Reasoning
CritPt
Real World Knowledge Work
GDPval-AA
Prompt Templates, Answer Extraction and Evaluation
Multiple Choice Questions (GPQ…
Multiple Choice Extraction Regex
Equality Checker LLM
HLE (Humanity's Last Exam)
HLE Equality Checker Prompt
LCR Equality Checker Prompt
Mathematical Questions (AIME 2…
Mathematical Equality Checker P…
Code Generation Tasks
SciCode

Artificial Analysis Intelligence Benchmarking Methodology

Artificial Analysis Intelligence Index v4.0.2 (January 2026)

Artificial Analysis Intelligence Index

Artificial Analysis Intelligence Index combines a comprehensive suite of evaluation datasets to assess language model capabilities across reasoning, knowledge, maths and programming.

It is a helpful synthesis of overall language model intelligence and can be used to compare language models. Like all evaluation metrics, it has limitations and may not apply directly to every use case. However, we are confident that it is a more useful synthesis comparison between language models than any other metric in existence today.

Artificial Analysis Intelligence Index Artificial Analysis Intelligence Index v4.0 incorporates 10 evaluations: GDPval-AA, τ^2-Bench Telecom, Terminal-Bench Hard, SciCode, AA-LCR, AA-Omniscience, IFBench, Humanity's Last Exam, GPQA Diamond, CritPt. Our methodology emphasizes fairness and real-world applicability.

We estimate a 95% confidence interval for Artificial Analysis Intelligence Index of less than ±1% - based on experiments with >10 repeats on certain models for all evaluation datasets included in Artificial Analysis Intelligence Index v4.0. Individual evaluation results may have wider than ±1% confidence intervals. We look forward to disclosing further detail from our statistical analysis in the future.

Artificial Analysis Intelligence Index is a text-only, English language evaluation suite. We benchmark models for image inputs, speech inputs and multilingual performance separately to the Intelligence Index evaluation suite.

Intelligence Index evaluation suite

Intelligence Index is calculated as a weighted average across four categories, each contributing 25% to the overall score. Category membership and weights are shown below.

Category	Evaluation	Field	Questions	Repeats	Response Type	Scoring	Intelli Index We
Agents (25%)	GDPval-AA	Real World Knowledge Work	220 tasks	1	Agentic task completion with file outputs	Pairwise comparison (ELO), frozen & scaled clamp((ELO-500)/2000)	16.
	τ^2-Bench Telecom	Agentic Workflows	114	3	Dual control agent-user simulation	World state evaluation, pass@1	8.3
Coding (25%)	Terminal-Bench Hard	Agentic Workflows	44	3	Terminal-based task execution	Test suite pass/fail, pass@1	16.
	SciCode	Code Generation	338 subproblems	3	Python Code (must pass all unit tests)	Code execution, pass@1, sub-problem scoring with scientist-annotated background prompting	8.3
General (25%)	AA-LCR	Long Context Reasoning	100	3	Open Answer	Equality Checker LLM, pass@1	6.2
	AA-Omniscience	Knowledge & Hallucination	6,000	1	Open Answer	50% Accuracy + 50% (1 - Hallucination Rate)	12.
	IFBench	Instruction Following	294	5	Open Answer	Extraction and rule-driven assessment, pass@1	6.2
Scientific Reasoning (25%)	HLE (Humanity's Last Exam)	Reasoning & Knowledge	2,158	1	Open Answer	Equality Checker LLM, pass@1	12.
	GPQA Diamond	Scientific Reasoning	198	5	Multiple Choice (4 options)	Regex extraction, pass@1	6.2

https://artificialanalysis.ai/methodology/intelligence-benchmarking

Issues with ML models

- Class imbalance: Disproportional number of samples per class.
  - Classifier might be biased towards majority classes and achieve bad classification.

- Concept drift: describe problem of changing underlying relationships in data.
  - Approximating a mapping function (f) given input data (x) to predict an output value (y), y = f(x).
    - Digit classification, text categorization or speech recognition, assume that mapping learning from historical data will be valid for new data and relationships between input and output do not change over time. Not true for the problem of malware detection and classification.

Issues with ML models

- Adversarial learning is a technique employed to attempt to fool machine learning by automatically crafting adversarial examples.

- Interpretability: Model is given an input X and it produces an output Y through a sequence of operations hardly understandable to a human.
  - Interpretability of the model determines how easily the analysts can manage and assess the quality and correct the operation of a given model.
    - Example: Cybersecurity analysts prefer rule-based and signature-based systems rather than neural-based methods

Handling ML experiments

- Who created the model at what time?
- Which hyperparameters were used?
- What feature transformations have been applied?
- Which dataset was the model derived from?
- Which dataset was used for computing the evaluation data?

Model Management Challenges

- Model Definition: Difficult to define actual model to manage.
  - Input data needs to be transformed into features expected by model.
  - ML pipelines combines feature transformations and actual model in a single abstraction.

- Model Validation: Ability to back-test accuracy performance of models over time (after model gets update).
  - Make use of same training, test and validation set including same evaluating metrics.

- Model Retraining: Deciding when to retrain a model is challenging.
- Training is done offline and models are loaded at prediction in a system.
- If new events (new public holiday, a new promotional activity) occur that our models have not been trained on, need to trigger retraining or even re-modelling.

Model Management Challenges

- Data Management Challenges: data integration, feature transformation, model training apply different operations based on different abstractions.
- Querying model Metadata to select best model: To accelerate model lifecycle management, need to understand metadata and lineage of models (e.g., hyperparameters, trained and validated datasets).
  - Centralized metadata store can accelerate learning processes (e.g., warm-starting of hyperparameter search) and automate simple error checks.
- Multi-Language Code Bases
- Backwards Compatibility: Model that was trained (last month or last year) should still be working today (required for production deployments).

Tips: Missing Value Handling

- Example: Missing values are problematic for product catalogs of online retailers.
- Manual update is not scalable.
- ML methods are designed for matrices only.
- Data is not available in numeric formats but rather in text/other forms.
  - Sols: Implementing feature extraction steps and imputation algorithms in one single pipeline.
- Must support hyperparameter optimization to automatically perform model selection and parameter tuning.

Summary

- Responsible AI (RAI) is a set of principles guiding the design, development, deployment, and use of AI to build trust in AI solutions.
- Aim of RAI is to embed ethical principles into AI applications.
- Pillars of Trust in RAI include explainability and interpretability, which are essential for trustworthy AI, encompassing prediction accuracy, traceability, and decision understanding.
- Fairness in RAI involves diverse and representative data, bias-less algorithms, and the establishment of an ethical AI review board.
- Robustness in RAI means handling exceptional conditions like input abnormalities and malicious attacks.
- Transparency in RAI refers to understanding how a service works, evaluating its functionality, and comprehending its strengths and limitations.
- Privacy in RAI is important, with regulatory frameworks like GDPR being relevant.
- Implementing responsible AI practices includes defining responsible AI principles, educating employees, integrating ethics across the AI development lifecycle, and protecting user privacy.

Summary

- Ethical issues in AI are a key concern.
- AI alignment is the process of encoding human values and goals into LLMs to make them helpful, safe, and reliable.
- Goal of AI alignment is to enable enterprises to tailor AI models to follow their business rules and policies. Alignment happens during fine-tuning, involving instruction-tuning and a critique phase.
- Importance of AI alignment stems from the increasing impact and risks of Deep Learning (DL) applications, with misalignment being a significant source of these risks.
- Misalignment has two main causes: reward hacking, where proxy rewards are optimized instead of actual rewards, and goal misgeneralization, where the agent pursues objectives different from training objectives.

Summary

- Formal verification methods like SAT, LP, and SMT are used for robust ML.
- Issues with ML models include class imbalance, concept drift, adversarial learning, and a lack of interpretability.
- Handling ML experiments involves tracking model creation details, hyperparameters, feature transformations, and datasets used.
- Model management challenges include defining the model, validation, retraining, and ensuring backwards compatibility.
- Data management challenges involve data integration, feature transformation, and querying model metadata.